The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
  1. Falling fertility (Guinnane 2011)
  2. Declining marriage (Stevenson and Wolfers 2007)
  3. Converging gender (income) gaps (Goldin 2014)
- Many policies targeting individual outcomes
- Existing research often treats them in isolation or pairwise, lacking a unified framework

This Paper

1. Document a three-way trade-off: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically unlikely

2. Develop a unified model of marriage, fertility, and female labor supply to explain the empirical facts

3. Show that redistributing childcare from wives to husbands can improve all three outcomes under certain conditions

4. Calibrate to Mexico (1990–2015), conduct decomposition and prediction

Key Findings

1. Traditional policies cannot resolve the trade-off; redistributing childcare to husbands can under two conditions
2. Mutual information analysis reveals synergy: the three outcomes must be analyzed jointly
3. Quantitative results on Mexico from 1990 to 2015:
   - → Gender-neutral TFP explains half of declining fertility and marriage
   - → Gender-biased TFP and education gap reversal drive income convergence
4. Dynamic propagation through gendered impacts of single parenthood

Literature

- Pairwise studies of fertility, marriage, and gender gaps (Galor & Weil 1996, Regalia & Ríos-Rull 2001, Greenwood et al. 2016)
  $\rightarrow$ Establish three-way trade-off; unified framework
- Gender-biased technical change (Greenwood et al. 2005, Ngai & Petrongolo 2017)
  $\rightarrow$ Show gender-neutral TFP also drives convergence
- Boys' disadvantage in single-parent families (Autor et al. 2019)
  $\rightarrow$ Embed in macro model with dynamic propagation
- Policy evaluation in family economics (Doepke & Kindermann 2019)
  $\rightarrow$ Identify policy frontier; childcare redistribution as remedy

Roadmap

- Empirical findings
- Model
- Policy analysis
- Quantitative analysis
- Dynamic extension
- Conclusion

Motivating Facts

Empirical Approach

1. Create panel datasets with fertility, dual parenthood, and gender income gap
2. Create dummy variables at sample medians:
   - → High fertility (F)
   - → Dual parenthood (M)
   - → Gender income equality (G)
3. Visualize intersections using Venn diagrams
4. Compute mutual information to characterize dependency structure

Dartboard approach: random benchmark is $0.5 \times 0.5 \times 0.5 = 12.5\%$

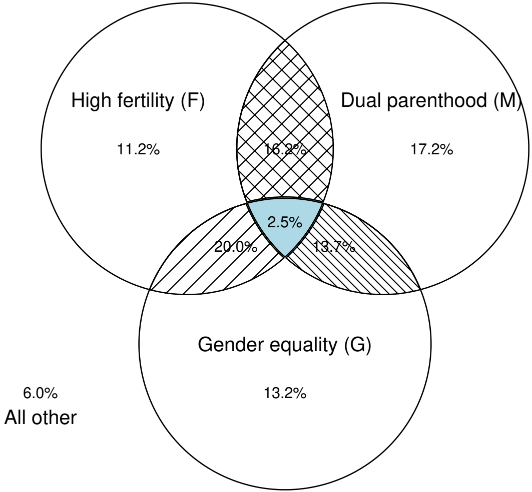
OECD Sample

37 OECD countries, 1970–2014

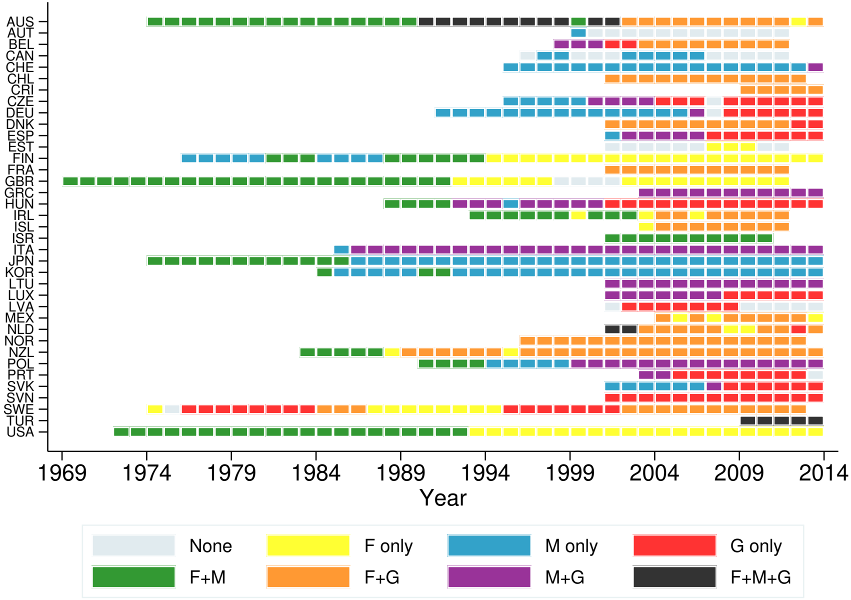
721 observations

- Total fertility rate
- Out-of-wedlock birth share
- Gender median earnings gap

Trinity share: 2.5%



OECD Sample – Transition Paths



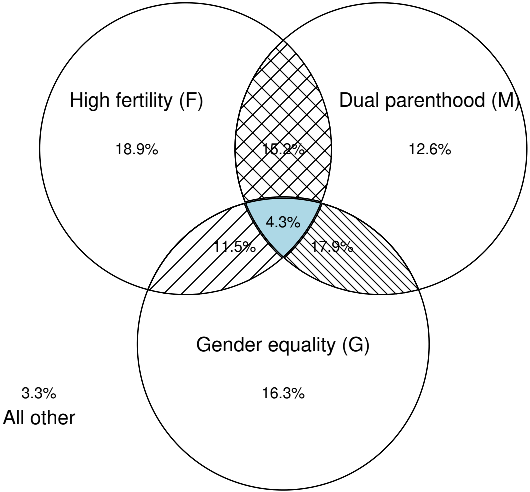
World Sample

95 countries, 1990–2015

1,130 observations

- Total fertility rate
- Children with both parents
- Female labor income share

Trinity share: 4.3%

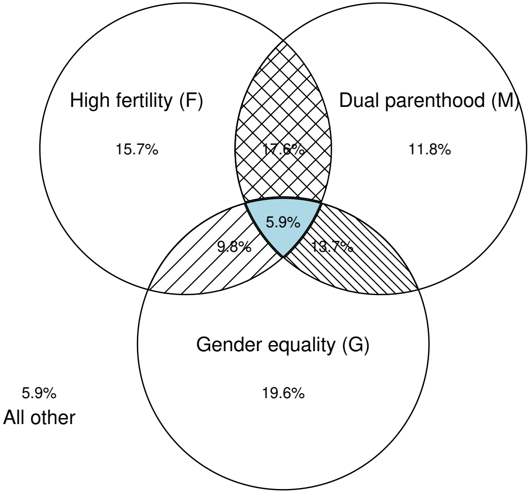


U.S. Sample (States)

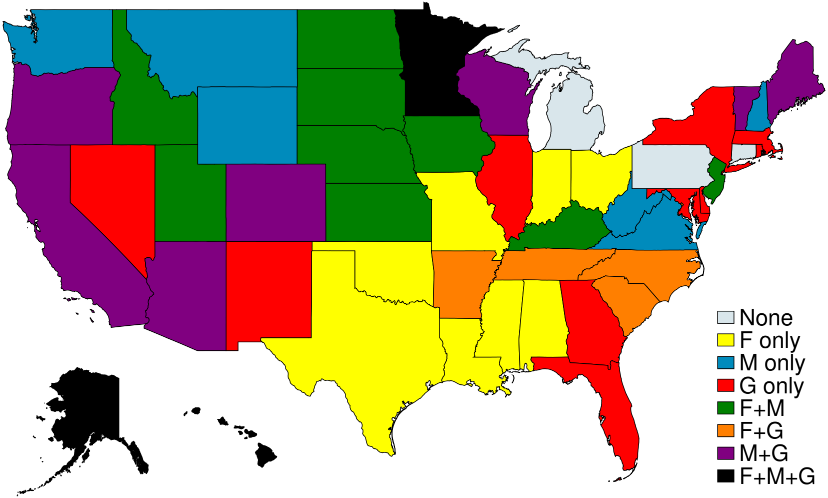
50 states + D.C., 2023

- Births per 1000 women
- Single-mother household share
- Gender earnings gap

Trinity share: 5.9%



U.S. Sample (States) – Geographic Distribution



Mutual Information Analysis

- Key finding: Synergy — dependence between any two variables *increases* when conditioning on the third
- Implication: analyzing pairs misses dependency structure

Sample	Total Corr.	Interaction	Interpretation
OECD	0.165	-0.035^{***}	Synergy
World	0.169	-0.026^{***}	Synergy
U.S. States	0.112	-0.007	Synergy

Questions

1. What explains the three-way trade-off?
2. Are there policies that can help escape the trade-off?
3. What role does technological progress play?

Model

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{♂, ♀\}$ and preference

$$u(c, n) = \left((1-\beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} \tag{1}$$

  where $\rho > 1$ so that $u(c, 0)$ is well-defined
- Homogeneous wage within gender: $w^{♂}$ and $w^{♀}$
- Gender wage gap: $\Gamma^w = w^{♂}/w^{♀}$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\male,s} = u(w^{\male}, 0) \tag{2}$$

- Once married, husbands transfer α share of income to wives

$$V^{\male,m} = u((1-\alpha)w^{\male}, n^m) \cdot \lambda \tag{3}$$

→ α is endogenous in marriage market equilibrium

→ λ is the psychic benefit of marriage

Marriage and Fertility – Women

- Single women solve

$$V^{\female,s} = \max_{c,l,n} \quad u(c,n) \quad \text{s.t.} \quad c = w^{\female} l, \quad l = 1 - \chi n \tag{4}$$

- Married women solve

$$V^{\female,m} = \max_{c,l,n} \quad u(c,n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\male} + w^{\female} l, \quad l = 1 - \chi n \tag{5}$$

- Fertility subject to wife's veto (Doepke & Kindermann 2019) $\implies$ wives determine marital fertility

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with scale θ
- Marriage rates:

$$\mathcal{M}^{\male} = \frac{1}{1 + (V^{\male,s}/V^{\male,m})^{\theta}}, \qquad \mathcal{M}^{\female} = \frac{1}{1 + (V^{\female,s}/V^{\female,m})^{\theta}} \tag{6}$$

- Equilibrium: $\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M}$
- (α, n^m) act as market-clearing “prices”

Aggregate Quantities

- Aggregate fertility and dual parenthood share:

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s, \qquad \mathcal{D} = \frac{\mathcal{M} \cdot n^m}{n} \tag{7}$$

- Female labor supply:

$$l^{♀} = \mathcal{M} \cdot l^m + (1 - \mathcal{M}) \cdot l^s = 1 - \chi n \tag{8}$$

- Gender income gap:

$$\Gamma^y = \frac{y^{♂}}{y^{♀}} = \frac{\Gamma^w}{l^{♀}} \tag{9}$$

Equilibrium Characterization

- Lemma 1: For any wage pair $\{w^{♂}, w^{♀}\}$, there exists a unique equilibrium (α, n^m)
- Lemma 2: Gains from marriage: $V^{♀,m}/V^{♀,s} = \lambda \cdot (1 + \alpha\Gamma^w)$
- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha\Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha\Gamma^w))^\theta} \tag{10}$$

$$\Gamma^y = \Gamma^w / l^{♀} \tag{11}$$

$$l^{♀} = 1 - \chi n \tag{12}$$

Three-Way Trade-Off: Intuition

- High n + High $\mathcal{M}$: Low $l^{♀}$ requires large Γ^w to sustain marriage $\Rightarrow$ high Γ^y
- High n + Low Γ^y: Low $l^{♀}$ requires tiny Γ^w for equality $\Rightarrow$ low $\mathcal{M}$
- High $\mathcal{M}$ + Low Γ^y: Large Γ^w needed for marriage; offsetting requires high $l^{♀}$ $\Rightarrow$ low n

Core mechanism: Women's disproportionate childcare burden ties fertility to gender income gaps through labor supply

Policy Analysis

Traditional Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	$+$	$+$	$+$ (widens)
Gender equality ($\uparrow w^{♀}$)	$-$	$-$	$-$ (narrows)
Reduce child costs ($\downarrow \chi$)	$+$	$+$	$+$ (widens)

Key insight: None of these policies can simultaneously improve all three outcomes

Redistributing Childcare to Husbands

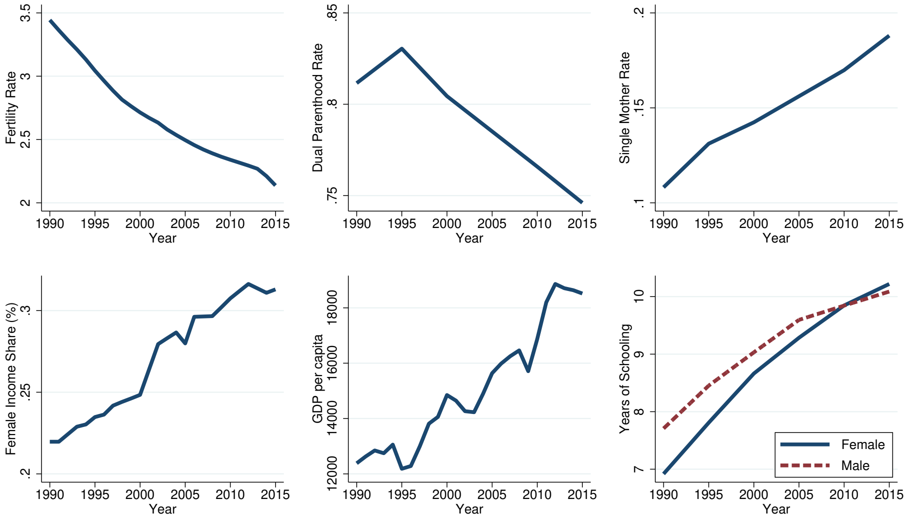
- Extend model: $\chi^{♀}$ for wives, $\chi^{♂}$ for husbands
- Wife's effective childcare cost: $\tilde{\chi}^{♀} = \frac{\alpha w^{♂} \chi^{♂} + w^{♀} \chi^{♀}}{\alpha w^{♂} + w^{♀}}$
- Proposition: Redistribution improves all three outcomes if:
  1. Wife's earnings exceed spousal transfers: $\alpha < w^{♀} / w^{♂}$
  2. Husband's utility gain from fertility exceeds consumption loss
- When both hold, redistribution is Pareto-improving

Policy Implications

- The trade-off may be partly an artifact of gendered childcare division
- Policies promoting paternal involvement (e.g., “use-it-or-lose-it” paternity leave) may help escape the trade-off
- Caveat: When wives’ earnings are low relative to transfers, this way out is not feasible
- Success depends on the economic environment and social norms

Quantitative Analysis

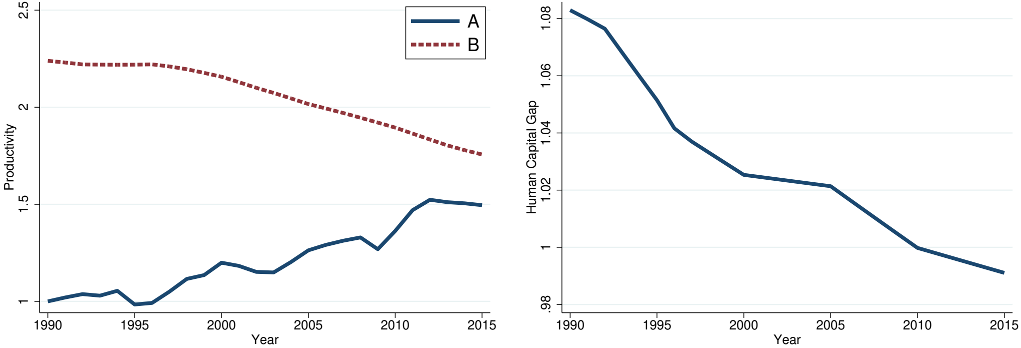
Mexico’s Transition (1990–2015)



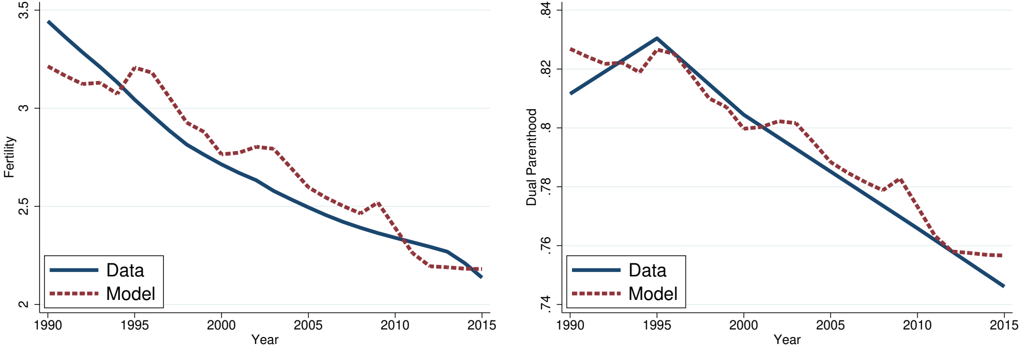
Calibration Strategy

- Decompose wages into three components:
  - $\rightarrow$ $A_t = w_t^{\male}$: gender-neutral productivity
  - $\rightarrow$ Γ_t^h: gender gap in human capital
  - $\rightarrow$ $B_t = (w_t^{\male}/w_t^{\female})/\Gamma_t^h$: gender-biased productivity
- Fix $\chi = 0.075$ (de La Croix and Doepke 2003)
- Calibrate: $\beta = 0.104$, $\rho = 1.5$, $\lambda = 1.03$, $\theta = 3.4$
- Key finding: $\alpha \approx 0.10$ satisfies condition for wives to benefit; husband condition also satisfied $\Rightarrow$ redistribution would be Pareto-improving

Calibration Results

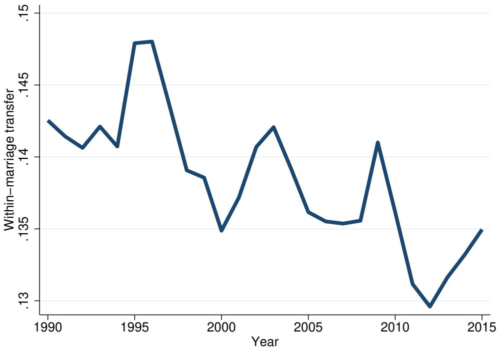


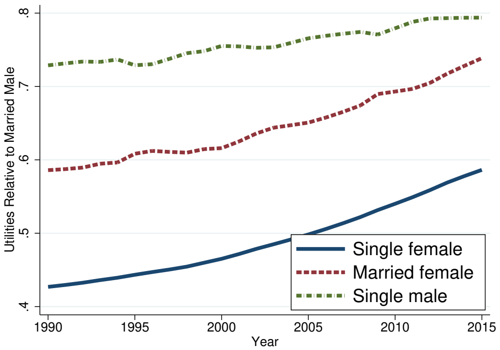
Model Fit



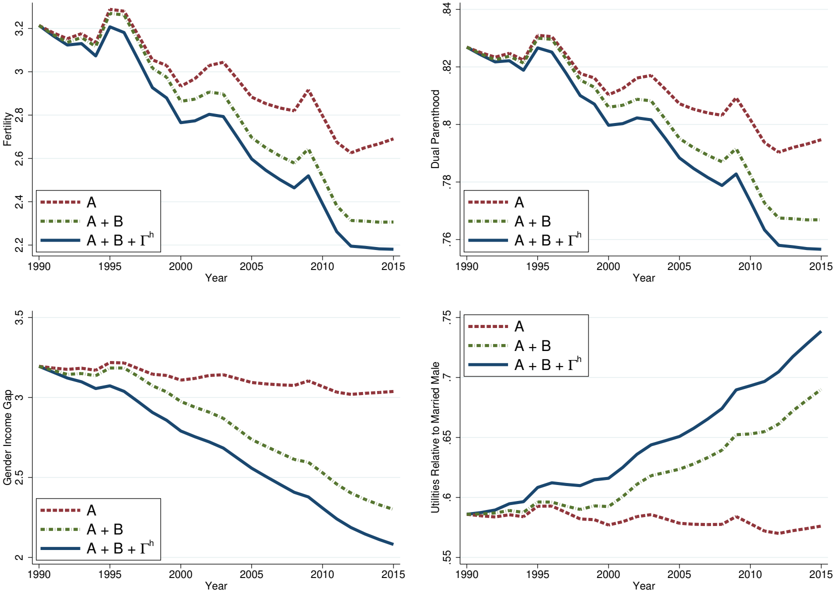
Fertility
3.5
3
2.5
2
1990
1995
2000
2005
2010
2015
Year
Data
Model
Dual Parenthood
.84
.82
.8
.78
.76
.74
1990
1995
2000
2005
2010
2015
Year
Data
Model

Equilibrium Outcomes





Decomposition



Decomposition Summary

- Gender-neutral productivity (A_t): $\sim$50% of fertility and marriage decline; 10% of income convergence
- Gender-biased productivity (B_t): $\sim$40% of fertility and marriage decline; majority of income and welfare convergence
- Human capital convergence (Γ_t^h): accounts for the remainder

Model-Based Prediction

Dynamic Extension

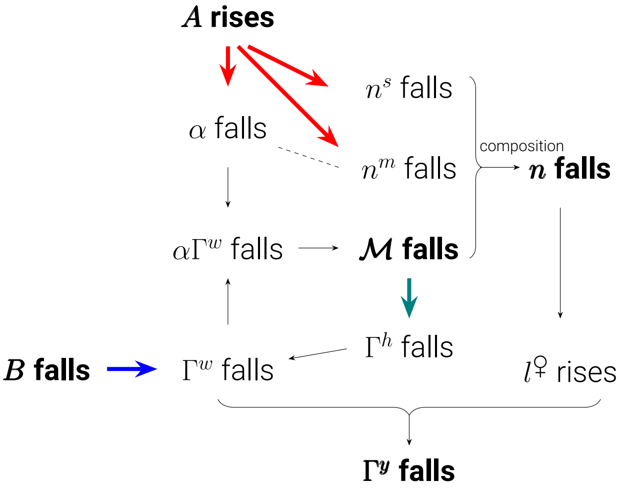
Endogenous Gender Human Capital Gap

- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022
- Law of motion:

$$\Gamma^h_{t+1} = \mathcal{H}(\mathcal{M}_t) \qquad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \tag{13}$$

- Creates self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline

Dynamic Mechanism



Red: A_t rises   Blue: B_t falls   Teal: Dynamic feedback

Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework explaining the empirical pattern
- Traditional policies cannot resolve the tension; redistributing childcare to husbands can under certain conditions
- Quantitative analysis: both gender-neutral and gender-biased technological change matter
- Dynamic propagation through human capital amplifies effects across generations